

Nitesh Shantha Kumar

Software Engineer · Distributed systems, data & reliability

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Software engineer who designs event-driven backend systems: Kafka pipelines, PostgreSQL that stays fast under load, and the reliability work that keeps both running. Technical lead on a retail pricing and promotions platform, and builder of agentic tooling for engineering teams. Background in robotics and safety-critical avionics software.

EXPERIENCE

Nordstrom

Seattle, WA · Aug 2021 – Present

Software Engineer — Pricing and Promotions Platform

- Technical lead for a clearance and promotions platform with REST and Kafka ingestion; split delivery into parallel workstreams and own design review and integration.
- Move legacy batch pipelines to event-driven publishing on Kafka with the transactional outbox pattern, using feature-flagged, shadow-mode cutovers with rollback plans.
- Design for scale and resilience: asynchronous processing, rate limiting, event-fed read stores, and fallbacks to upstream services.
- Diagnose database and throughput problems, lead incident investigations, and write root-cause analyses with prioritised follow-ups.
- Build alerting and runbooks, plus agentic tooling (skills and subagents) that automates incident triage, used across teams.

Ak Aerotek

Bengaluru, India · May 2016 – May 2019

Software Engineer

- Verification and validation of safety-critical avionics software to DO-178B/C, writing Ada test harnesses and running them on target hardware.
- Led a small team through code and audit reviews.

SKILLS

Languages	Java, Python, SQL, C++
Distributed systems	Apache Kafka (Avro, log compaction, ACLs, Confluent Cloud), transactional outbox, event-driven and service-oriented design, REST APIs, concurrency, fault tolerance
Data	PostgreSQL (query plans, indexing, MVCC), Oracle, DynamoDB, Neo4j, Redis / Valkey, Flyway
Cloud & operations	AWS (EKS, S3, RDS, Aurora), Kubernetes, Docker, New Relic, Splunk, JMeter, incident response and on-call
AI & tooling	Agents, skills and subagents, Model Context Protocol (MCP), LangGraph, retrieval over code and knowledge graphs

SELECTED PROJECTS

- **Universal MCP Manager** — Electron app for installing and managing MCP servers across Claude Code, VS Code and Cursor.
- **AlgoMentor** — local-first interview-prep desktop app with a Socratic tutor running on local Claude Code or Codex CLIs.
- **Neo4jAI / SqlAI** — natural-language interfaces to Neo4j and PostgreSQL through LangGraph agents and MCP servers.
- **Bug2 & FireBot** — robot motion planning and a ROS firefighting robot from graduate robotics work.

EDUCATION

M.S., Engineering Sciences (Robotics) — University at Buffalo, NY

2019 – 2021

Graduate Research Assistant, autonomous-vehicle perception and sensor fusion (LANDR Lab)

B.E., Electronics & Communication — Visvesvaraya Technological University

2011 – 2015